AgriVision: Product Overview

What AgriVision Is

AgriVision is software that helps farmers find crop problems early and treat only the areas that need it. It studies video from a drone flight, spots insect pests, plant diseases, and stressed crops, and shows exactly where they are on a map of the field.

From there, it works out how much treatment each affected area needs and prepares a plan for a spraying drone. A person always reviews and approves the plan before anything is applied.

The Problem It Solves

Most farms spray an entire field even when only small patches are affected.

As a result, much of the chemical lands on healthy crops or bare soil; the project analysis cites estimates as high as 70–90%. That waste raises costs, can pollute nearby water, and helps pests grow resistant. Meanwhile, inspecting fields on foot is slow, so problems are often found only after yield has been lost.

How It Works

AgriVision turns a drone flight into a treatment plan in five steps.

1. **Fly:** A drone flies over the field and records video along with its GPS position data.
2. **Detect:** Artificial intelligence reviews the footage and flags pests, diseases, and stressed crops.
3. **Map:** Each finding is placed at its real-world location. Repeated sightings are confirmed, and one-off false alarms are filtered out.
4. **Plan:** The software outlines the affected areas, divides them into 33-foot (10-meter) squares, and calculates the treatment needed for each square.
5. **Approve and export:** A person reviews and signs off on the plan. AgriVision then produces a flight-plan file for a spraying drone, plus map files.

What It Can Detect

AgriVision looks at fields in two ways, depending on how close the camera is.

Up close, it recognizes about 130 types of insect pests and leaf diseases across crops such as corn, tomato, potato, grape, apple, and bell pepper. From the air, it looks for field-wide patterns: weed patches, nutrient shortages, drought stress, standing water, storm damage, and planting gaps. It also checks leaf color and texture to catch yellowing, browning, and wilting.

Flight height matters. At about 33 feet (10 meters), individual leaf spots are visible. At about 100 feet (30 meters), only broad patterns can be seen.

What You Get

Each analysis produces results a farm manager can act on right away.

Output	What it gives you
Annotated video	Detected problems, marked and color-coded by severity
Field map	Infected areas and the proposed treatment squares
Treatment plan	The amount of treatment for each square, so nothing is over- or under-applied
Export files	A flight-plan file for compatible spraying drones, plus map files for tools such as Google Earth

The result is less chemical used, lower costs, earlier detection, and a documented record of every approval.

Where It Stands Today

AgriVision is a working software prototype that analyzes footage after a flight.

Ready today	Still to come
Analyzes recorded drone video on a laptop	Live video analysis during flight
Maps findings and calculates treatment amounts	A direct connection to the drone
Exports flight plans and map files; all automated tests pass	Control of the spray equipment itself

One practical note: standard GPS can be off by about 8 feet, so the drones need high-precision positioning to reach the right squares. The software plans the job, and a separate spraying drone carries it out.

Recommended Path Forward

The recommended approach is a three-stage workflow: scout, approve, then spray.

First, a lightweight drone scouts the field at 33–50 feet; about 20 minutes covers roughly 37–50 acres. Next, AgriVision runs on a field laptop, and an agronomist reviews and signs off on the plan. Finally, the approved flight plan goes to a dedicated spraying drone, which treats only the marked squares.

This approach works with the software as it exists today, keeps chemicals away from sensitive electronics, and keeps a human in control. The next step is a field trial on a real farm, using high-precision drones and an experienced spraying-drone operator.